

The empirical recurrence for OEIS A300927, recovered from its tail

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Abstract

OEIS A300927 records “Empirical recurrence of order 64” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 88$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 68$. Its last coefficient is $c_{64} = -2 \neq 0$, so no further tail satisfies anything shorter; any order-64 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A300927 is “Number of $n \times 5$ 0..1 arrays with every element equal to 1, 2, 3, 4 or 5 horizontally, vertically or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

3, 129, 3447, 87561, 2244335, 57594572, 1477281166, 37892542998, ...

The entry states:

Empirical recurrence of order 64 (see link above)

As of the “Last modified” line on the live entry (Mar 15 2018 18:59:09, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 88$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 88$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 64: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 64.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 176$ exact terms from $n = 4$ on, it returns order 64 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{64} c_i a(n-i),$$

c_1	24	c_{17}	732043184	c_{33}	−1335440603	c_{49}	18126633
c_2	76	c_{18}	−4268731314	c_{34}	−13489636535	c_{50}	40839217
c_3	−724	c_{19}	−2512759482	c_{35}	−13305919878	c_{51}	20778711
c_4	−3875	c_{20}	8558513313	c_{36}	−12799296895	c_{52}	7707941
c_5	4240	c_{21}	871613478	c_{37}	−9086416957	c_{53}	269445
c_6	79148	c_{22}	−19176002026	c_{38}	−1173461193	c_{54}	−666907
c_7	136299	c_{23}	−775218790	c_{39}	359600340	c_{55}	−310622
c_8	−687849	c_{24}	24363302121	c_{40}	3964156388	c_{56}	−157603
c_9	−2938864	c_{25}	5528232974	c_{41}	3469002052	c_{57}	−30845
c_{10}	827618	c_{26}	−3251080247	c_{42}	2265828045	c_{58}	−15730
c_{11}	25676169	c_{27}	7663167457	c_{43}	1190093403	c_{59}	−3005
c_{12}	36097608	c_{28}	2629577617	c_{44}	−278415340	c_{60}	719
c_{13}	−108299526	c_{29}	12701932724	c_{45}	−383706287	c_{61}	366
c_{14}	−347911716	c_{30}	26486006651	c_{46}	−532764680	c_{62}	186
c_{15}	126495408	c_{31}	17786012631	c_{47}	−247875138	c_{63}	17
c_{16}	1603489270	c_{32}	10327559666	c_{48}	−82298137	c_{64}	−2

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 68$, and it is the recurrence OEIS A300927 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{64} - c_1 x^{64-1} - \dots - c_{64}$. Its constant term is $-c_{64} = 2$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 64 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 64, so $\deg q = 64$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 64, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 64, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -2 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-64 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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